

Bryanna Elizabeth Binion

AI/ML Software Engineer

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PROFESSIONAL EXPERIENCE

Senior AI Full-Stack Engineer

Apr 2021 - Present

PreScouter

- Played a key role in back-end development and prompt engineering, leveraging advanced technologies such as Python, Whisper, and ElevenLabs to optimize system functionality and performance.
- Developed and fine-tuned a well-trained LLM model using real therapy data within the psychotherapist chatbot to enhance its conversational capabilities in addressing mental health concerns effectively.
- Implemented and integrated text-to-speech, speech-to-text, and text-to-text functionalities in real-time streaming, enabling seamless verbal and text interactions with users for providing therapy and guidance.
- Enhanced the Chatbot's memory capabilities to store and recall previous conversations, ensuring personalized and coherent interactions with users during subsequent sessions.
- Leveraged LangChain and LangGraph infrastructures to simplify modifications to the AI model, while utilizing LLaMa 3.1 hosted on AWS Bedrock to ensure comprehensive user privacy control.
- Migrated to AWS SageMaker for complete control over model training and deployment, while addressing privacy concerns through a custom LangChain class for streaming outputs, without altering LangGraph workflows.
- Developed APIs and services using Flask for deploying machine learning models, enabling real-time predictive capabilities.
- Implemented voice modulation for inputs and outputs, using a faster-whisper solution on AWS SageMaker for privacy, while leveraging the ElevenLabs API for outputs.
- Created custom memory logic using LangGraph to track conversation flow and reference past therapy sessions, maintaining an updated client profile.
- Built and improved data pipelines for machine learning workflows using Apache Spark and SQL, enabling efficient model training and real-time data processing.
- Engineered cloud-based solutions on AWS, utilizing services such as S3, EC2, Lambda, and SageMaker to deploy scalable and reliable machine learning workflows.
- Explored MLOps strategies by integrating Kubeflow pipelines for end-to-end model lifecycle management, enabling reproducibility and scalability in deployment.
- Achieved test coverage exceeding 95%, ensuring high quality and reliability in the codebase.
- Integrated PostgreSQL and AWS DynamoDB into the workflow for this functionality.
- Collaborated with the project team to develop a user-friendly front-end interface, facilitating easy access and navigation for users seeking mental health support.
- Maintained the frontend application with React 17, continuously adding new features using Bootstrap integration.
- Implemented OpsGenie alerts and DataDog metrics, enabling quick and well-informed decision-making.
- Technologies Used : Python, Flask, React 17, LLM, PostgreSQL, Docker, Kubernetes, EKS

Machine Learning - Backend Engineer

Jan 2016 - Mar 2021

Talent.com | *Chicago, IL*

- Developed HR Helper Bot on Telegram assists HR professionals by filtering job candidates based on specified criteria and preferences.
- Processed 2.3M Job Descriptions using fuzzy text matching algorithms and saved \$100K automating manual efforts.
- Built a recommendation engine using Sentence Transformers and Redis, increasing customer engagement by 40%.
- Performed extensive data preprocessing and quality assurance using Pandas and NumPy, ensuring data consistency, integrity, and readiness for analysis and model training.
- Implemented secure authentication using OAuth2 and JWT for AI service endpoints.
- Introduced automated ML pipelines with MLflow, streamlining workflows from model experimentation to deployment.
- Collaborated effectively with product, design, and research teams to align AI functionalities with user requirements and product goals, fostering synergy and ensuring successful integration of AI capabilities.
- Leveraged microservice integration and deployment techniques alongside an inference manager, achieving a 15% improvement in recognition accuracy and a 20% reduction in processing time.
- Maintained and enhanced AI systems in production for reliability, accuracy, and robustness while utilizing New Relic for comprehensive monitoring to ensure real-time visibility into system performance.
- Technologies Used : Python, Pandas, Numpy, Redis, MySQL, New Relic

Data Scientist

Aug 2019 - Dec 2020

Ecolab | *Kennesaw, GA*

- Collaborated with teams to build ML model for system stress prediction of projected cost savings of \$1M.
- Utilized VIF scores to detect multicollinearity and systematically eliminating redundant variables.
- Developed guidelines for data preparation for predictive model Crude flex (prediction of oil quality).
- Automated sales performance reports with Advance Excel and VBA, saving 100 monthly manhours.
- Designed PowerBI dashboards with DAX queries to track KPIs, sustaining a 95% customer satisfaction rate.
- Technologies Used : Python, TensorFlow, StatsModels, Pandas, Scikit-learn, VBA, DAX

Education

Master's Degree in Computer Science, University of Georgia

06/2017 - 07/2019

Bachelor's Degree in Computer Science, University of Georgia (GPA : 3.84)
Specialization in **Algorithms and AI/ML**

10/2013 - 05/2017